

Data Sources and Processing Pipeline

This study integrates three heterogeneous data sources capturing complementary aspects of individual-level traits and behaviors: (i) personality questionnaire data based on the Big Five (OCEAN) framework, (ii) behavioral outcomes from laboratory-based economic games implemented via the MobLab platform, and (iii) value orientation data from the World Values Survey (WVS).

The processing procedures for the Big Five and Moblab datasets followed publicly available pipelines adopted in prior research [1, 2]. The personality questionnaire and value orientation data were obtained from publicly available sources, while the laboratory-based economic game behavioral data are made publicly available through our repository. The processing of the World Values Survey dataset follows the official document on “constructing indices for Secular and Emancipative Values”¹.

Across all datasets, processing was guided by three core principles: (1) filtering out invalid or incomplete entries at the sample level, (2) retaining only features directly relevant to the theoretical constructs, and (3) transforming and aggregating item-level responses into interpretable, continuous indices following established measurement practices or clearly defined computational rules. Below we describe further details in processing the data. The code for data processing are provided in our repository.

1 Personality Data Processing

Personality data were collected using a standardized Big Five questionnaire, with each dimension measured by multiple five-point Likert-scale items. During processing, all item responses except for the country column were converted to integer values to ensure numerical consistency, and samples with invalid or missing responses on key items (e.g., E1) were filtered out to guarantee the reliability of subsequent analyses.

Feature selection was restricted to items corresponding to the five personality dimensions (O, C, E, A, N), with reverse-scored items identified a priori and transformed accordingly. Item-level responses within each dimension were then summed to produce five continuous trait scores (score_O, score_C, score_E, score_A, score_N), effectively aggregating item-level data into interpretable indices.

While the processing and scoring pipeline was implemented by the authors, it follows the canonical Big Five measurement procedures widely adopted in personality psychology, ensuring consistency across participants and experimental settings.

2 MobLab Behavioral Data Processing

Behavioral data from laboratory-based economic games were released through our repository. The raw dataset (joint.csv) was loaded and key columns were renamed for clarity and consistency across experiments.

To ensure reliable analysis, samples with missing values for each game were filtered out using a column-specific dropna procedure. The resulting cleaned columns were directly used as features for subsequent analyses of individual-level behavioral traits and decision-making patterns.

3 WVS Data Processing

Individual-level value orientation data were obtained from the World Values Survey (Wave 7). A subset of theoretically relevant questionnaire items was selected for subsequent analyses, including items related to autonomy (Q8, Q11, Q17), equality (Q33, Q29, Q30), choice (Q182, Q184, Q185), and voice (Q152, Q153, Q154, Q155).

Item-level responses were processed into four distinct indices, e.g., autonomy, equality, choice, and voice, using explicitly defined scoring rules that handle missing or inconsistent responses with predefined imputation and weighting schemes. These four indices were then aggregated into a single continuous index (evi_index) to represent overall value orientation, following established computational procedures widely used in prior WVS research.

To ensure comparability with other datasets, only observations with valid indices (evi_index ≥ 0) were retained for downstream analysis. A scaled score (score_scaled) was created by multiplying evi_index by 100 and rounding to the nearest integer. All processing and scoring code is publicly available in our repository.

References

- [1] Qiaozhu Mei, Yutong Xie, Walter Yuan, and Matthew O Jackson. A turing test of whether ai chatbots are behaviorally similar to humans. *Proceedings of the National Academy of Sciences*, 121(9):e2313925121, 2024.
- [2] Yutong Xie, Qiaozhu Mei, Walter Yuan, and Matthew O Jackson. Using large language models to categorize strategic situations and decipher motivations behind human behaviors. *Proceedings of the National Academy of Sciences*, 122(35):e2512075122, 2025.

¹World Values Survey’s guide on constructing indices for Secular and Emancipative Values: <https://www.worldvaluessurvey.org/WVSContents.jsp?CMSID=welzelidx>, accessed Feb 08, 2026.